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Assessment Tool-1

1. City hospital Data Warehouse

a. Snowflake Schema: A central Consultation-Fact table is connected to four dimension tables: Patient, Doctor, Department and Time. To make it a snowflake schema, these dimensions are normalized

b. OLAP Operations:

1. Roll-up on time from the base level to year
2. Roll-up on patient and doctor to all

Resulting view:

[department, year]

c. Cuboid Count: The total number of cuboids is the product of the levels in each dimension. With 4 dimensions and 4 levels each:

$$4 \times 4 \times 4 \times 4 = 256 \text{ cuboids}$$

2. Smart City Project

Fact and Dimension Design

* Dimensions: Time, Location, Data-Source

* Measures: vehicle count, avg-speed, air-quality-index, energy-kwh, ridership-count

* Justification: A star schema with these elements directly answers the project's analytical

needs. The measures quantify congestion, and efficiency, while the dimensions provide the necessary filtering contexts.

3. Banking Data Warehouse

a. Galaxy Schema: Two central fact tables (Transaction - fact and Loan - fact) share common dimension tables (Customer, Branch, Time and Account - Type).

b. OLAP operations:

1. Roll-up on Time to year

2. Roll-up on Customer and Account - Type to all
Resulting view groups data by [branch, year].

c. Cuboid Count: Since both fact tables share the exact same 4 dimensions with 4 levels each, the dimensional space for each cube is $4^4 = 256$.

4. Manufacturing Company:

* Roll-up: Aggregate data up the hierarchy by grouping product to category, rolling up factory to All, and filtering Time for the current year.

* Drill-down: Slice for a specific factory, then step down the hierarchy from factory to department, and Product to type.

* Slice:- Filter the cube for a single condition:
Supplier rating ≥ 4 and rollup Location across
all regions.

* Dice: Filter the cube on two or more
dimensions simultaneously.

ex:- Category IN(A,B) and Location IN(X,Y)

5. Retail Chain:

* Roll-up: Aggregate Time to quarter, apply a Slice
for "last quarter", rollup Product to Category,
and roll up Store to All

* Drill-down: Slice for a specific Region and the
Electronics category, then step down the product
hierarchy to subcategory and SKU

* Slice: Filter the cube for specific single-dimension
criteria: Gender = Male and Tier = Bronze, rolling
up other non-required dimensions.

* Dice: Select a sub-cube by applying multiple
filters: City IN(List) AND category IN(List) AND
Time IN(Specific Dates).